Lab 1: Urban Camping

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A) Dashboard: Urban Camping Restrictions in Missoula

- For our dashboard, we included a list of all the urban camping restrictions in the city of Missoula, as well as two different indicators. The first indicator shows the total city area that allows urban camping, and the second indicator shows the total city area that does *not* allow urban camping.
- If we had more time, the next steps would be to add more elements such as a live weather forecast, links to additional city resources (i.e. Houseless-Resource-Guide), and mapped emergency services (i.e. hospitals) to add to the usefulness for this resource.
- We originally wanted to add a live weather forecast indicator into our dashboard; we tried looking for ways to incorporate this data but were unsuccessful given current level of coding knowledge. While one of the other groups successfully incorporated live weather data, we chose to stick with indicators that illustrate the total square milage of camping and non-camping areas within the city.

B) Static Map Layout

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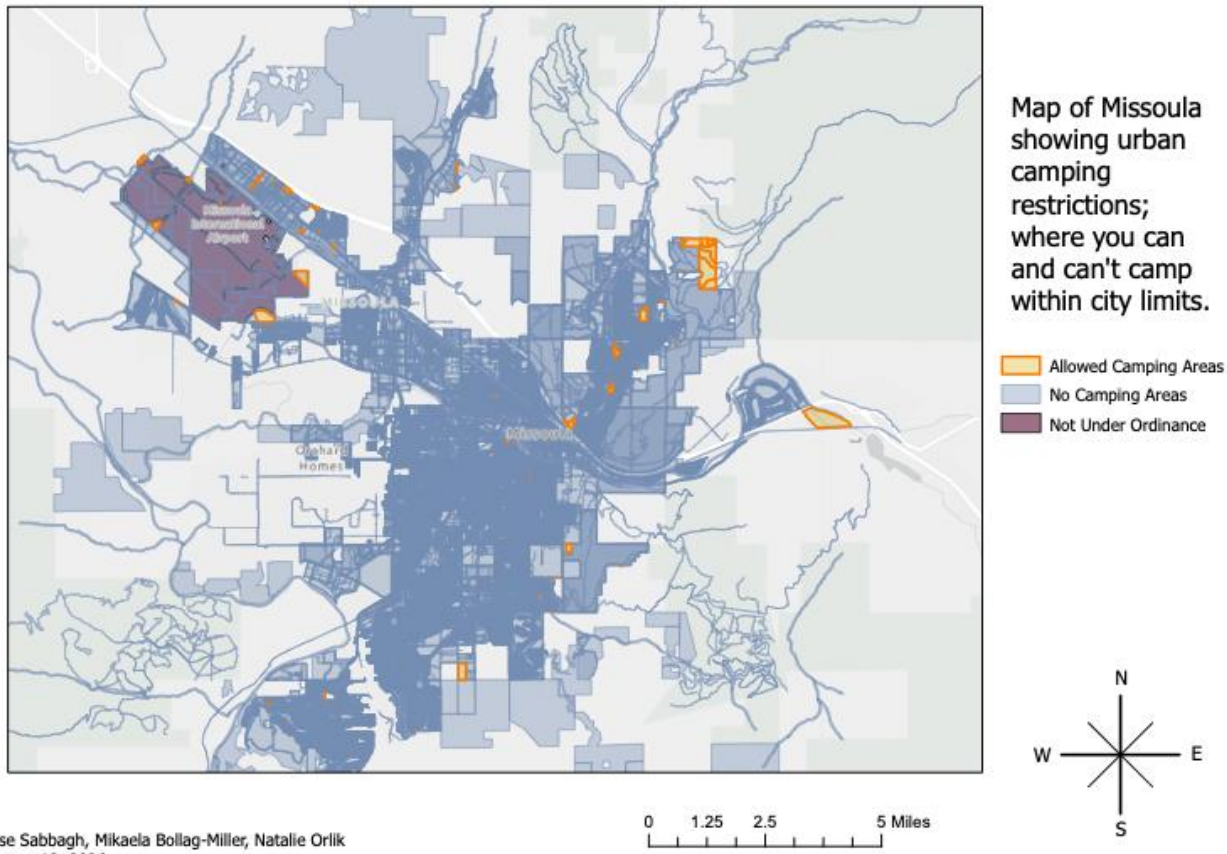
Urban Camping Restrictions in Missoula



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Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

- **Additional Information: Methods and group interpretation**

a. **Within 100 feet (approximately the length of a regulation basketball court) of a structure containing a residential dwelling unit (as defined by the 2021 International Building Code, as amended and adopted by the Montana Department of Labor and Industry).**

i. We interpret this to mean within 100 feet of private land polygons which include residential dwelling units and this also overlapped with our commercial business requirement (b.)

b. **Within 100 feet of a public entrance of any commercial business or non-profit organization.**

i. We interpret this to mean within 100 feet of the private land layer, as explained in a.; of which includes commercial business or non-profit organizations.

c. **Within 300 feet (approximately the length of a standard City block) of any parcel or lot containing any public or private school, including secondary, elementary, or preschool, or any lot or parcel containing a daycare.**

i. We interpret this to mean within 300 feet of the center point of any parcel or lot containing any public or private school, including secondary, elementary, or preschool. These buffer zones created round circles around buildings of relevance.

d. **Within 300 feet of any shelter facility (e.g. those providing shelter to unhoused populations, victims of domestic violence, transitional housing, etc.)**

i. We interpret this to mean within 300 feet of the center point of any shelter facility.

e. **Within 300 feet of an area designated by the Mayor as safe parking sites or designated temporary camping sites.**

i. No safe parking sites or designated temporary camping sites exist at this time of research.

f. **Within conservations lands regulated by Chapter 12.40Links to an external site.**

i. We interpret this as within conservation easements, legal agreements which restrict the type and amount of development on a property for conservation purposes (conservation easement | U.S. Fish & Wildlife Service)

g. **Within City parks regulated by Chapter 12.40Links to an external site.**

i. We interpret this as Parks owned and maintained by the City of Missoula

h. **On or within 10 feet of a bike lane, public pathway, trail, or trailhead, or so as to block free passage on a sidewalk (sidewalks must have six feet of clear, unobstructed space).**

i. We interpret this as within 10 feet of the centerline of a bike lane, public pathway (road), trail, or sidewalk. We assume trail heads are incorporated by trail layer.

i. **Within 50 feet of the high water mark for the Clark Fork River, Bitterroot River, Rattlesnake Creek, or Grant Creek, including any side channels or backwaters thereof, where those waterways are located within City limits.**

i. We interpret this as within 65 feet of the centerline of all rivers within City limits. In this interpretation, all high watermarks are likely to fall within 15 feet of the centerline of the river (30 ft being the maximum river width at high water).

j. **Areas temporarily or permanently signed or designated on the City's urban camping webpage as no camping due to environmental degradation or because of public health, safety and or welfare concerns.**

i. No designated no camping areas were noted on the City's urban camping webpage at this time of research.

Workflow for final analysis of all buffer layers:

- I merged the private land layer with the shelter buffer layer and the school buffer layer.
- I then took my other previously merged layers including parks, conservation lands, public lands, as well as a merged layer of rivers, trails, streets, etc., of which were also all buffered.
- Then I merged all these layers together
- Then, utilizing the erase tool, I subtracted all of these factors (restricted camping areas) from a layer of Missoula City boundaries.
- After this there were over 250 polygons! Wow, too many were wrong! --> utilizing the Multipart feature to Single part feature I was able to split the polygons into their own shapes, allowing me to select, edit, and delete them individually.
- After deleting over half of the polygons that were wrongfully placed (i.e., on houses/buildings/streams/etc.) I was left with around 60 features visualizing the available urban areas